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Ballistic Exercise Versus Heavy Resistance Exercise Protocols: Which Resistance Priming Is More Effective for Improving Neuromuscular Performance on the Following Day?

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Abstract

Nishioka, T and Okada, J. Ballistic exercise versus heavy resistance exercise protocols: which resistance priming is more effective for improving neuromuscular performance on the following day? *J Strength Cond Res* 37(10): 1939–1946, 2023—This study aimed to determine whether ballistic exercise priming (BEP) or heavy resistance priming (HRP) is more effective for improving ballistic performance after 24 hours. Ten resistance-trained men performed BEP and HRP conditions 72–144 hours apart in a randomized and counterbalanced order. Jumping performance was assessed before and 24 hours after the BEP and HRP sessions using 0 and 40% one-repetition maximum (1RM) squat jump (SJ), 0 and 40% 1RM countermovement jump (CMJ), and drop jump (DJ) reactive strength index (RSI). Statistical significance was accepted at $p \leq 0.05$. In the BEP condition, 0% 1RM CMJ height (+3.62%) as well as theoretical maximum velocity (+5.14%) and theoretical maximum power (+2.55%) obtained from CMJ 24 hours after the priming session were significantly greater than those at the baseline ($p \leq 0.05$), but 0% 1RM SJ height and DJ RSI ($p > 0.05$) were not greater than those at the baseline. In the HRP condition, the jump performances were not improved ($p > 0.05$). The percentage change in 0% 1RM CMJ height in the BEP condition was significantly greater than that seen in the HRP condition ($p = 0.015$) but did not differ for 0% 1RM SJ height and DJ RSI ($p > 0.05$). These results suggest that the BEP is more effective than HRP in improving CMJ performance after 24 hours. Therefore, practitioners should consider prescribing resistance priming using low-load ballistic exercises rather than high-load traditional exercises when planning to enhance athlete performance on the following day.

Key Words: priming exercise, delayed potentiation, jump performance, force-velocity profiling, power

Introduction

To elicit adaptations that enhance physical performance, repeated exposure to specific training challenges is required (39); however, single bouts of exercise can also produce a positive transitory performance response (36). Low-volume exercise stimuli have been demonstrated to improve short-term neuromuscular performance, with postactivation performance enhancement (PAPE) being the term used to describe this process (30). Although acute changes in neuromuscular characteristics have been demonstrated, the period in which performance outcomes can be positively altered is relatively short (e.g., 3–18.5 minutes) (2,35). As a result, PAPE methods may be difficult to implement in field settings (9). Nonetheless, recent studies (7,9) have suggested that these strategies may provide a delayed neuromuscular potentiation effect after extended durations of rest and have demonstrated that a session of very low–volume exercise stimulus (i.e., resistance priming) can increase a range of subsequent physical capacities for up to 48 hours.

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Priming exercise strategy factors, such as load and type of exercise, are essential in determining postpriming response (9,22). Harrison et al. (9) suggested that either low-load (30–40% one-repetition maximum [1RM]) ballistic exercise or high-load (≥85% 1RM) traditional exercise would be the most effective resistance priming. In this study, the former was termed as "ballistic exercise priming" (BEP), and the latter as "heavy resistance priming" (HRP). These priming exercises are often performed on the day of (e.g., 6 hours before) or the day before (e.g., 24 hours before) a competition (9,22). For instance, Saez Saez de Villarreal et al. (33) examined the effects of BEP and HRP on neuromuscular performance after 6 hours and found that both BEP and HRP significantly enhanced jump performance. Nevertheless, limited research (7,27,42) has investigated BEP and HRP, which was implemented to improve performance after 24 hours. Although a few previous studies (27,42) suggested that BEP was effective in improving jump performance after 24 hours, to the best of our knowledge, only 1 study (7) by Harrison et al. has examined the change in neuromuscular performance 24 hours after HRP. Furthermore, because no studies have directly compared the delayed potentiation effects of HRP and BEP on neuromuscular performance after 24 hours, it remains unclear which resistance priming is more effective when used on the day before a competition. It has been suggested that priming activity may have to be specific to subsequent performance requirements to maximize neuromuscular performance (9,22). If so, BEP rather than HRP may be a more effective priming strategy for improving ballistic performance, which is for the primary goal of many athletes. Such information can help practitioners better understand priming strategies for enhancing athletic performance and decide what type of resistance priming should be prescribed.

Therefore, this study aimed to determine whether BEP or HRP is more effective for improving ballistic performance after 24 hours. We hypothesized (a) that both BEP and HRP would improve neuromuscular performance on the following day and (b) that BEP would induce superior performance enhancements compared with HRP.

Methods

Experimental Approach to the Problem

A repeated-measures design with a randomized and counterbalanced order was used to investigate the delayed potentiation effects of resistance priming sessions on jumping performance after 24 hours. Subjects performed 2 sessions (BEP and HRP) 72–144 hours apart. To evaluate the ballistic performances, the following jump performances were assessed before and 24 hours after the BEP and HRP conditions: unloaded squat jump (SJ), unloaded countermovement jump (CMJ), and reactive strength index (RSI) from drop jump (DJ). In addition, for force-velocity profiling (Table 1), the loaded SJ and the loaded CMJ were also performed. High levels of reliability for the performance variables measured in this study were previously reported by our laboratory (27). Furthermore, because the primary purpose of this study was to compare the effects of 2 different resistance priming sessions, we did not include a control condition. The experimental design and procedures for the BEP and HRP conditions are detailed in Figure 1.

Subjects

Ten resistance-trained men aged between 19 and 25 years volunteered to participate in this study (age: 21.9 ± 1.8 years, height: 170.0 ± 4.3 cm, body mass [BM]: 69.6 ± 7.4 kg, half squat 1RM: 149.5 ± 16.8 kg, mean ± *SD*). These subjects had 13.4 ± 2.9 years of sports training background, 4.2 ± 1.7 years of resistance training experience, and were free of musculoskeletal pain or injury that could compromise testing. To compare the effects of BEP and HRP in individuals who could be more likely to experience performance enhancement using resistance priming (27), those who had sufficient lower-body strength (i.e., half squat 1RM > 1.90 × BM) were recruited. After explaining the study's purpose, procedures, risks, and benefits to the potential subjects, written informed consent was obtained before participation. To minimize confounding factors, instructions related to sleep and diet were provided to the subjects before the experiment was conducted. On the night preceding each test session, the subjects were asked to keep their usual sleeping habits, with a minimum of 7 hours of sleep. During the period of investigation (i.e., immediately before the experimental session), the subjects were instructed to avoid any known stimulants (e.g., caffeine) or depressants (e.g., alcohol) that could possibly enhance or compromise wakefulness. In addition, the subjects were requested to maintain their habitual physical activity and avoid strenuous activity the day prior, and throughout the study. This study was approved by the Ethics Review Committee on Human Research of the Waseda University (Approval number: 2020-368).

Procedures

Familiarization and Preliminary Measurements. During the preliminary session, i.e., 72–144 hours before the BEP or HRP conditions, the subjects were familiarized with the SJ (unloaded and loaded), CMJ (unloaded and loaded), DJ from 20-, 30-, and 40-cm heights, and jump squat with 40% estimated 1RM. In addition, the subjects performed 3 repetitions of DJ from 20, 30, and 40 cm with adequate (60–90 seconds) rest to determine the optimum drop height for the primary trials. The optimum drop height was defined as that with the greatest RSI (i.e., DJ height or ground contact time). This approach was selected based on available literature, which stressed the importance of identifying individual optimum drop heights to maximize neuromuscular adaptations (32). Accordingly, the average optimum drop height was 33.0 ± 9.0 cm. After 3 minutes of rest, the half squat 1RM of each subject was measured. The maximum strength of the lower body was measured using a knee angle of 90° because half squat strength strongly correlated with jump performance (45). The half squat 1RM testing was performed by having the subjects

Table 1

Definitions and practical interpretations of the force-velocity profile variables in squat jump and countermovement jump (24).

Variable	Definition and computation	Practical interpretation
$\overline{F_0}$ (N·kg^{-1})	Theoretical maximal force production of the lower limbs as extrapolated from the linear loaded jump squats' force-velocity relationship; *y*-intercept of the linear force-velocity relationship	Maximal concentric force output (per unit body mass) that the athlete's lower limbs can theoretically produce during ballistic push-off. Determined from the entire force-velocity spectrum, it gives more integrative information on force capability than, e.g., concentric squat 1-repetition maximum load
$\overline{v_0}$ (m·s^{-1})	Theoretical maximal extension velocity of the lower limbs as extrapolated from the linear loaded jump squats' force-velocity relationship; *x*-intercept of the linear force-velocity relationship	Maximal extension velocity of the athlete's lower limbs during ballistic push-off. Determined from the entire force-velocity spectrum and very difficult, if not impossible, to reach and measure experimentally. It also represents the capability to produce force at very high extension velocities
$\bar{P}_{max}$ (W·kg^{-1})	Maximal mechanical power output, computed as $\bar{P}_{max} = \overline{F_0} \times \overline{v_0}/4$ or as the apex of the power-velocity 2nd-degree polynomial relationship	Maximal power output capability of the athlete's lower-limb neuromuscular system (per unit body mass) in the concentric and ballistic extension motion
S_{Fv} (N·s·kg^{-1}·m^{-1})	Slope of the linear force-velocity relationship, computed as $S_{Fv} = -\overline{F_0}/\overline{v_0}$	Index of the athlete's individual balance between force and velocity capabilities. The steeper the slope, the more negative its value, the more "force-oriented" the force-velocity profile, and vice versa

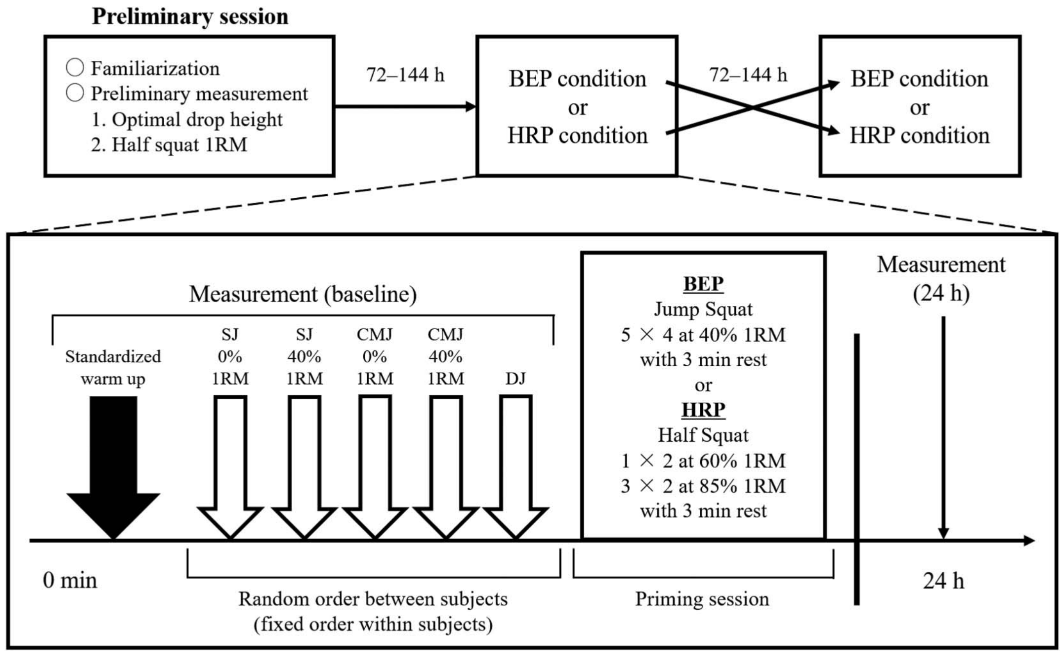

Figure 1. Experimental design and procedures involved in ballistic exercise priming (BEP) and heavy resistance priming (HRP) conditions. SJ = squat jump; CMJ = countermovement jump; DJ = drop jump; RM = repetition maximum.

complete a series of warm-up sets (5 repetitions at 30% estimated 1RM, 3 repetitions at 50% estimated 1RM, 2 repetitions at 70% estimated 1RM, and 1 repetition at 90% estimated 1RM), each separated by a 3-minute recovery period, after which a series of maximal lift attempts were performed until a 1RM was obtained. The subjects performed the downward movement of the half squat exercise until reaching the lowest point, which was determined by a beep sound activated when a photocell beam was interrupted by the posterior portion of the thigh when a knee angle of 90° was reached. The knee angle was measured at the lowest point of movement of the half squat using a smartphone video camera set at 240 Hz (iPhone 12, Apple Inc., Cupertino, CA). Two-dimensional motion analysis of the data was performed using Kinovea Video Analysis Software (ver. 0.8.15, www.kinovea.org). The knee angle was calculated by digitizing reflection markers, which were attached to the greater trochanter, lateral epicondyle of the femur, and lateral malleolus. The line connecting the greater trochanter with the lateral epicondyle of the femur, and the line connecting the lateral epicondyle of the femur with the lateral malleolus, created an angle, which was used to define the knee angle. Trials in which the subjects did not reach a knee angle of less than 90° of flexion were excluded from the analysis.

Measurement of Squat Jump, Countermovement Jump, and Drop Jump Performance. For the SJ and the CMJ, the subjects held a 0.1-kg plastic bar for the 0% 1RM assessment and a 20-kg barbell loaded with the appropriate weight plates for the 40% 1RM assessment. During the SJ, the subjects were instructed to lower into a half squat position with the desired knee angle, which was indicated by the beep sound, as discussed previously. They then held the position for 2 seconds and then jumped as high as possible when instructed while performing no previous countermovements. The waveform data obtained from the force plate during the trial was checked to confirm that no previous countermovement was used. For the CMJ, the subjects were instructed to perform the downward movement as deep as the SJ, the countermovement as fast as possible, and the jump as high as possible. For the DJ, the subjects were asked to step off a wooden box at a set height without lifting their center of gravity and then land on the force plate with both feet. In addition, they were instructed to rebound and immediately jump as high as possible after contact while minimizing ground contact time. Their hands were kept akimbo throughout the jump, although a straight body position during landing and take-off was encouraged. They were also required to land back on the force plate.

Ballistic Exercise Priming and Heavy Resistance Priming Conditions. The subjects performed a standardized warm-up that included 3 minutes of jogging, 10 minutes of dynamic stretching (43), and 2 repetitions each of the SJ and the CMJ at submaximal effort (i.e., approximately 80% of maximal effort). This was followed by 2 repetitions each of SJ and CMJ with 0% 1RM, 20 kg, 30% 1RM, and 40% 1RM with adequate (60–90 seconds) rest periods. After completing the standardized warm-up, the subjects performed 3 repetitions each of the SJ and the CMJ at 0% 1RM with 90 seconds of rest, 3 repetitions each of the SJ and the CMJ at 40% 1RM with 2 minutes of rest, and 3 repetitions of the DJ with 90 seconds of rest. The order of these jump performance measurements was randomized for each subject and maintained for all subsequent measurement sessions. After an additional 3 minutes of rest, the subjects performed the priming sessions (BEP: 5 sets of jump squats for 4 continuous repetitions per set and 40% 1RM with 3 minutes of rest; HRP: 1 set of half squats for 2 repetitions per set and 60% 1RM followed by 3 sets of half squats for 2 repetitions per set and 85% 1RM with 3 minutes of rest). Because the difference in BEP and HRP exercises (i.e., jump squat vs. half squat) made perfectly matching total volume loads difficult, the priming protocols recommended in a previous study (9) were used in this study. The subjects were instructed to perform the jump squat and half squat exercises with a countermovement to a knee angle of 90° and to push the ground "as hard and fast as possible" during the descent and ascent phases. However, in the HRP session, the subjects were instructed to prevent the barbell from having momentary flight time after the ascent phase. The subjects were not allowed to jump after the ascent phase. The subjects' jump performance was measured 24 hours after the priming session. All jump performance measurements were made at the same time of day, between 14:30 and 19:00 hours, to avoid diurnal variations. To prevent dehydration, ad libitum drinking was permitted, and environmental conditions were controlled (18° C–22° C and 20–60% humidity) during all testing sessions.

To quantify the internal load of each priming session (i.e., session rating of perceived exertion [RPE]), 15 minutes after the end of BEP or HRP, the subjects were asked, "How was your priming workout?" Subjects answered the question according to the CR-10 scale (15,38,40). They were instructed to choose a descriptor and its respective number from 0 to 10 (0 = rest; 1 = very easy; 2 = easy; 3 = moderate; 4 = somewhat hard; 5–6 = hard; 7–9 = very hard; and 10 = maximum) (38,40).

Visual Analog Scales. The subjects completed visual analog scales (VAS, 100 mm scale) to record perceptions of fatigue and muscle soreness during the BEP and HRP sessions (27,41). Immediately after the subjects arrived, a VAS for fatigue anchored with verbal descriptors ranging from "not fatigued at all" to "extremely fatigued" was provided to each subject who was asked to rate their general feeling of "fatigue and tiredness." A VAS for muscle soreness anchored with verbal descriptors ranging from "no soreness" to "extremely sore" on which subjects were asked to rate their "muscle soreness and pain throughout their entire lower extremities during performing CMJ with bodyweight" immediately after they performed 2 repetitions of CMJ at 0% 1RM during the standardized warm-up was collected.

Measurement Equipment and Data Analyses. All jumps (SJ, CMJ, and DJ) were performed on a single force platform (0625, ACP, AccuPower; AMTI, Watertown, MA) that sampled vertical ground reaction force (GRF) data at a frequency of 1,000 Hz using an analog-to-digital converter (EIRBZ22002369; CONTEC Co. Ltd., Osaka, Japan) and subsequently recorded them to a personal computer. Signals from the force plate were filtered by a 50 Hz low-pass, zero-phase-lag finite impulse response filter. Before each jump evaluating SJ and CMJ, the subjects were weighed over 3 seconds with the external load laid on their shoulders to determine the total system weight (sum of body mass and external mass). The start of the jump was defined as the time point 30 milliseconds before the vertical GRF exceeded the threshold (the total system weight ± 5 *SD*) (28). For each jump, the system's center of mass (COM) velocity was calculated using the trapezoid rule (17), whereas the net GRF was calculated as the amount of force exceeding the system weight divided by the system mass to determine acceleration. Acceleration was numerically integrated to provide instantaneous COM velocity, which in turn was numerically integrated to provide instantaneous COM displacement. The beginning of the concentric phase was identified as the instant at which the vertical COM velocity exceeded 0 $m{\cdot}s^{-1}$ after the start of the downward phase. The end of the concentric phase was identified as the instant at which the vertical GRF decreased to <20 N after the beginning of the concentric phase (10). For DJ, the contact and take-off were defined as the time points at which the vertical GRF exceeded or decreased below the threshold (i.e., 20 N), respectively. The DJ height was determined using the flight time method (26,42).

For the force-velocity profiling of the SJ and CMJ, the mean vertical force developed by the lower limbs during push-off (i.e., concentric phase) and the corresponding mean COM vertical velocity were determined using the equations validated by Samozino et al. (34) and Jiménez-Reyes et al. (11). The total system weight, push-off distance, and jump height data substituted for Samozino's equations (34) were derived from the vertical GRF. The push-off distance was identified as the distance of the vertical COM displacement from the start to the end of the concentric phase. The jump height was calculated using the take-off velocity procedure (17). The force and velocity data obtained under 2 different loads (0 and 40% 1RM) were modeled using a least-squares linear regression model to determine the force-velocity profile: $\bar{F}(\bar{v}) = \overline{F_0} - a{\cdot}\bar{v}$, where $\overline{F_0}$ represented the theoretical maximum force (i.e., force intercept) and $\overline{v_0}$ was the theoretical maximum velocity (i.e., velocity intercept) corresponding to the slope of the linear force-velocity relationship ($S_{Fv} = -\overline{F_0}/\overline{v_0}$) (24). The average push-off distance during 0 and 40% 1RM was used in the analyses. The theoretical maximum power ($\bar{P}_{max}$) was calculated as $\bar{P}_{max} = \overline{F_0}{\cdot}\overline{v_0}/4$. This 2-point method was used to minimize stimuli and fatigue during performance testing based on distant loads and was validated by García-Ramos et al. (5) as a quick and less fatigue-inducing procedure for testing the forcevelocity profile.

All force and power values were normalized to the subject's body mass. Average values over 3 jump repetitions were analyzed because average CMJ height provides better sensitivity for monitoring neuromuscular status compared with the highest CMJ height (3).

Statistical Analyses

Normality was tested using the Shapiro–Wilk test. For pairwise comparisons, paired-samples *t*-tests were used, or nonparametric Wilcoxon tests when normal distributions were not observed. Effect size was determined by Cohen's *d* for parametric data (small: 0.20–0.49, medium: 0.50–0.79, large: >0.8) and Cliff delta (19) for nonparametric data (small: 0.15–0.32, medium: 0.33–0.46, large: >0.46). In addition, the smallest worthwhile change was calculated using 0.2 × between-subject *SD* (7) to identify individual changes in jump variables. "Positive or negative responding" and "nonresponding" individuals were identified when the individual change was smaller or greater than the smallest worthwhile change in each variable. Values were presented as mean ± *SD*. Statistical analyses were performed using SPSS statistical software (SPSS Statistics, Ver. 27; IBM, Armonk, NY), with $p \leq 0.05$ indicating statistical significance.

Results

Baseline values for all performance variables were not significantly different between BEP and HRP conditions ($p > 0.05$). In the BEP condition, 0% 1RM CMJ heights (+3.62%, +0.013 m, 95% confidence interval [CI]: 0.006–0.019 m, $p = 0.002$, $d = 0.201$), CMJ $\overline{v_0}$ (+5.14%, +0.138 $m{\cdot}s^{-1}$, 95% CI: 0.066–0.211 $m{\cdot}s^{-1}$, $p = 0.009$, Cliff delta = 0.220), and CMJ $\bar{P}_{max}$ (+2.55%, +0.600 $W{\cdot}kg^{-1}$, 95% CI: 0.338–0.863 $W{\cdot}kg^{-1}$, $p = 0.009$, Cliff delta = 0.160) 24 hours after the priming session were greater than those at the baseline (Figure 2, Table 2). In addition, CMJ S_{Fv} 24 hours after the BEP session was lower than that at the baseline (−6.77%, +0.951 $N{\cdot}s{\cdot}kg^{-1}{\cdot}m^{-1}$, 95% CI: 0.240–1.662 $N{\cdot}s{\cdot}kg^{-1}{\cdot}m^{-1}$, $p = 0.028$, $d = 0.376$) (Table 2). However, the HRP

Table 2

Changes in the force-velocity profile variables during squat jump (SJ) and countermovement jump (CMJ).*†

		BEP		HRP	
Jump	Variable	Baseline	24 h	Baseline	24 h
SJ	$\overline{F_0}$ (N·kg^{-1})	35.13 ± 3.22	34.43 ± 3.17	35.52 ± 3.09	35.14 ± 3.08
	$\overline{v_0}$ (m·s^{-1})	2.76 ± 0.79	2.86 ± 0.80	2.71 ± 0.46	2.67 ± 0.29
	$\bar{P}_{max}$ (W·kg^{-1})	24.00 ± 5.66	24.38 ± 5.59	23.97 ± 4.29	23.41 ± 3.37
	S_{Fv} (N·s·kg^{-1}·m^{-1})	−13.47 ± 3.40	−12.71 ± 3.00	−13.49 ± 2.70	−13.33 ± 1.86
CMJ	$\overline{F_0}$ (N·kg^{-1})	35.09 ± 3.21	34.21 ± 2.62	34.90 ± 3.02	34.53 ± 4.36
	$\overline{v_0}$ (m·s^{-1})	2.84 ± 0.60	2.98 ± 0.60‡	2.88 ± 0.51	2.90 ± 0.64
	$\bar{P}_{max}$ (W·kg^{-1})	24.78 ± 4.98	25.38 ± 4.96‡	25.07 ± 4.68	24.91 ± 5.34
	S_{Fv} (N·s·kg^{-1}·m^{-1})	−12.83 ± 2.72	−11.88 ± 2.32‡	−12.46 ± 2.36	−12.39 ± 3.02

*$\overline{F_0}$ = theoretical maximum force; $\overline{v_0}$ = theoretical maximum velocity; $\bar{P}_{max}$ = theoretical maximum power ($\overline{F_0}\cdot\overline{v_0}/4$); S_{Fv} = slope of the force-velocity relationship ($-\overline{F_0}/\overline{v_0}$); BEP = ballistic exercise priming.

†Values are presented as mean ± *SD*.

‡Significantly different from baseline in the BEP condition ($p \leq 0.05$).

session did not significantly improve jump performance ($p > 0.05$). Squat jump performance (Figure 2 and Table 2) and DJ RSI (Figure 2) as well as DJ height and ground contact time did not change significantly under the BEP and HRP conditions ($p > 0.05$). In percentage change analyses, 0% 1RM CMJ height in the BEP condition showed greater improvement than in the HRP condition (+5.62%, 95% CI: 1.22–10.02%, $p = 0.015$, $d = 1.418$), with no differences observed in other variables between the BEP and HRP conditions ($p > 0.05$).

Baseline values for perceptions of fatigue and muscle soreness were not significantly different between the BEP and HRP conditions ($p > 0.05$). In the BEP and HRP conditions, perceptions of muscle soreness (+11.4 mm, 95% CI: 3.08–19.72 mm, $p = 0.031$, Cliff delta = 0.600 and +15.4 mm, 95% CI: 4.01–26.79 mm, $p = 0.033$, $d = 1.122$, respectively) 24 hours after the priming sessions were significantly greater than that observed in the baseline (Table 3), although perceptions of fatigue were not increased 24 hours after the priming sessions in both conditions ($p > 0.05$). Session RPE for the BEP and HRP conditions was similar (3.70 ± 0.78 and 3.70 ± 0.64, $p = 1.000$).

Discussion

This study aimed primarily to examine whether BEP or HRP was more effective in improving ballistic performance after 24 hours. The primary findings of this study suggest that BEP effectively enhanced CMJ performances (i.e., unloaded CMJ height, CMJ $\overline{v_0}$, and CMJ $\bar{P}_{max}$) after 24 hours. By contrast, no significant effect on jump performance the following day was observed for HRP. In addition, the improvement in unloaded CMJ height with BEP was significantly greater than that with HRP. Although these findings supported the hypotheses that BEP is effective for improving neuromuscular performance the following day and that its effect was greater than HRP, the hypothesis that HRP is also effective for enhancing neuromuscular performance after 24 hours was not confirmed.

This study showed that BEP specifically improved CMJ performances (i.e., 0% 1RM CMJ height, CMJ $\overline{v_0}$, and CMJ $\bar{P}_{max}$) on the following day (Figure 2 and Table 2), whereas SJ performance, DJ RSI, and CMJ F0 (i.e., force output ability at low velocities) did not improve. The jump squat exercise performed as the BEP seems more similar to CMJ in using the stretch-shortening cycle and a high contribution of hip extension torque (1,44). In addition, because BEP was performed using a low load, the movement velocity during the priming exercise was relatively high (23). Therefore, the BEP may have been a more specific and effective priming stimulus for the force output ability at high rather than low velocities (12). This is consistent with the results of previous studies (27,42). Based on these findings, the results of this study can provide additional strong evidence for the hypothesis that delayed potentiation occurs in a movement-specific and velocity-specific manner associated with priming exercises. Furthermore, because the degree of improvement in unloaded CMJ height associated with the BEP condition was greater than that for the HRP condition (Figure 2), BEP can be considered a more effective priming strategy for improving neuromuscular performance on the following day.

It is noteworthy that when individual responses for the BEP condition with 0% 1RM CMJ height were evaluated, none of the subjects had a negative response (Figure 2). The BEP session was characterized by priming exercises performed with maximal movement intent but with low external loads (i.e., 40% 1RM). Linnamo et al. (16) suggested that heavy resistance loading may result in greater fatigue of central and peripheral origin with reduced muscle electrical activity accompanied by an accumulation of blood lactate than explosive and lighter loading. In addition, using lower external loads may reduce muscle damage induced by performing exercises (31). Based on these observations, BEP seems to be an appropriate priming strategy for minimizing negative factors that impair subsequent neuromuscular performance.

This is the first study to investigate the effect of HRP performed in the afternoon on neuromuscular performance after 24 hours. Harrison et al. (7) reported that high-load priming (i.e., 1 × 3 at 67% 1RM, 1 × 3 at 77% 1RM, and 2 × 2 at 87% 1RM) performed in the morning did not improve unloaded SJ and CMJ height after 24 hours. Because this study also did not find any performance improvement 24 hours after HRP, HRP may not be effective for improving performance after 24 hours, no matter when it is performed during the day. Although physical performance, such as maximum strength, has been reported to be higher in the afternoon than in the morning (13), given these findings, this diurnal variation may not significantly affect the stimulus or benefits obtained from priming exercises.

Considering that BEP improved jump performance, although HRP did not, may be because of the movement characteristics of each priming exercise. Ballistic exercise priming is characterized by the intention to perform a movement at maximal velocity and by the acceleration of a mass throughout an entire movement (21,25). Such ballistic movements allow for greater force output

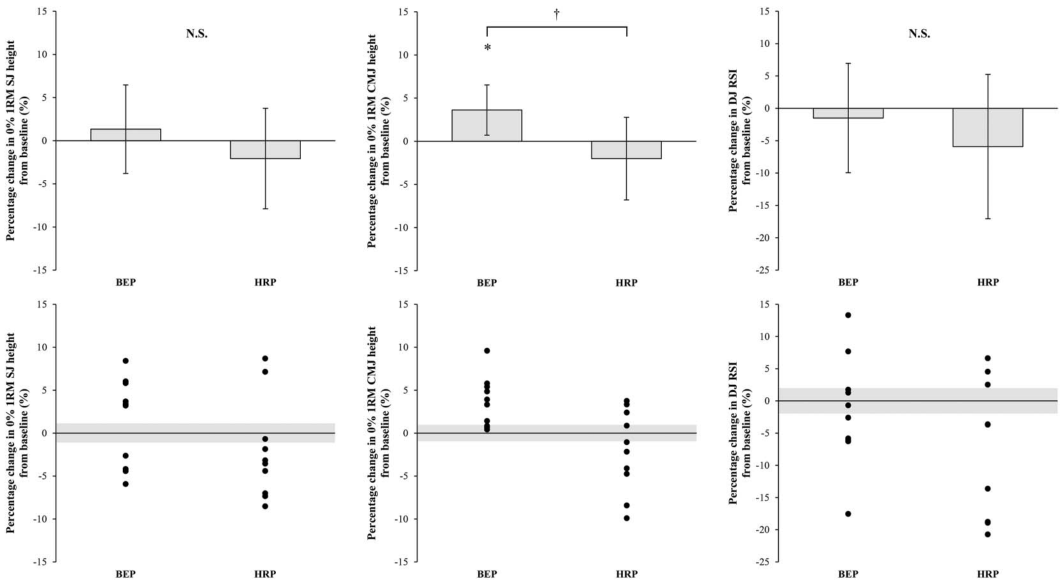

Figure 2. Percentage changes (mean ± *SD*) compared with corresponding baseline value in 0% one-repetition maximum (1RM) squat jump (SJ) height, 0% 1RM countermovement jump (CMJ) height, and drop jump (DJ) reactive strength index (RSI). Individual data for each metric are annotated along the bottom. The shaded areas represent the smallest worthwhile change (0.2 × between-individuals *SD*). *$p \leq 0.05$ from the corresponding baseline value; †$p \leq 0.05$ between ballistic exercise priming (BEP) and heavy resistance priming (HRP) conditions; N.S. = not significant.

and muscle excitation during resistance exercises (25). However, in HRP performed with a nonballistic exercise (i.e., back squat), a deceleration phase occurs at the end of the exercise movement (14). During this deceleration phase in resistance exercises, when no more force is exerted than the system mass, the recruitment of motor units throughout the movement may not be maximized (14,21,25). The presence of this deceleration phase may attenuate the priming stimulus, which is important for improving jump performance.

The lack of performance improvement in the HRP condition might also be explained by potential muscle damage. In fact, HRP induced an increase in the perception of muscle soreness after 24 hours (Table 3). Previous studies (28,30) suggested that higher-intensity resistance exercises with eccentric contractions induce greater muscle damage. Both HRP and BEP were exercised with eccentric contractions, but HRP was performed at higher intensities than BEP (60–85% 1RM vs. 40% 1RM, respectively). Therefore, using high loads during priming exercises (i.e., HRP) may increase potential muscle damage and inhibit performance enhancement. However, it is also true that there have been previous studies that suggested that HRP may be an effective priming exercise. For example, Cook et al. (4) reported that performing 3RM bench presses and squats in the morning improved CMJ peak power, 40-m sprint time, and 3RM bench press and squat performance in the afternoon of the same day. In addition, González-García et al. (6) found that 80% 1RM half squats performed in the morning significantly improved CMJ height in the afternoon of the same day. Although other studies have indicated that HRP is an effective priming strategy, they all showed benefits to neuromuscular performance in the afternoon of the same day (e.g., after 6 hours) (4,6,9,22). Based on the above findings and the results of this study, it is likely that HRP is effective when implemented in the morning on the day of competition, whereas it may not be effective when performed the day prior. This may be explained by the characteristics of the performance recovery process following exercise. Raastad and Hallén (31) observed that SJ performance dropped immediately (i.e., within 5 minutes) after performing high-intensity resistance exercise (e.g., 3RM squats), followed by recovering in the afternoon, with a second drop occurring after 22 hours. After eccentric exercise of the quadriceps muscle, infiltration of leukocytes has been reported to gradually increase and peak in concentration 20–25 hours after exercise (20). Furthermore, it has been suggested that phagocytic infiltration 24 hours after exercise is associated with increased protein degradation in mice (18). Such physiological mechanisms may be related to the observed lack of performance improvement 24 hours after HRP. Future studies will need to investigate how HRP affects

Table 3

Changes in the perceptions of fatigue and muscle soreness measured using visual analog scales (100-mm scale).*†

	BEP		HRP	
	Baseline	24 h	Baseline	24 h
Fatigue (mm)	12.9 ± 8.9	16.7 ± 9.3	15.6 ± 14.7	21.0 ± 10.6
Muscle soreness (mm)	9.4 ± 12.8	20.8 ± 13.8‡	12.0 ± 7.6	27.4 ± 17.9§

*BEP = ballistic exercise priming; HRP = heavy resistance priming.
†Values are presented as mean ± *SD*.
‡Significantly different from baseline in the BEP condition ($p \leq 0.05$).
§Significantly different from baseline in the HRP condition ($p \leq 0.05$).

subsequent neuromuscular performance, including physiological markers.

Interestingly, session RPE for the BEP and HRP were similar, which indicates that the overall exercise intensities (≈ workload × sets × reps × rep time per rest time) (15) for both protocols were roughly equivalent. Several studies (15,29,37) have shown that higher overall exercise intensity induced greater muscle damage. This assumes that the lack of performance gains associated with HRP is not because of differences in overall exercise intensity. Furthermore, given the fact that BEP significantly improved CMJ performance, it seems that the overall exercise intensity of HRP was appropriate as a priming stimulus. Indeed, both the HRP and the BEP did not significantly increase the perception of fatigue (Table 3). Nevertheless, HRP was not effective for improving neuromuscular performance after 24 hours. This suggests that the strategy of using high loads itself may not be optimal when resistance priming is performed to improve neuromuscular performance on the following day. In addition, session RPE for the BEP and HRP performed in this study were similar to the overall exercise intensity actually prescribed the day before a competition by practitioners who used priming strategies (8). Therefore, the results of this study will have significant and practical implications for coaches and athletes in the field.

Practical Applications

These results suggest that BEP is an effective strategy for improving CMJ performances (i.e., unloaded CMJ height, CMJ $\overline{v_0}$, and CMJ $\bar{P}_{\max}$) after 24 hours. In particular, to enhance unloaded CMJ height, BEP can be a more effective resistance priming strategy than HRP. Furthermore, HRP may not be effective for improving neuromuscular performance after 24 hours. Therefore, practitioners should consider prescribing resistance priming using low-load ballistic exercises rather than high-load traditional exercises when planning to enhance athlete performance on the following day.

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